



Measuring and appraising placebo effects in clinical trials: contemporary challenges and approaches in psychiatry

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Placebo effects are among the greatest paradoxes in medical research. Although more data on placebos are available from placebo-controlled trials than on any treatment in medicine, there has been comparatively little deep interrogation of issues related to their measurement, appraisal, and interpretation. Most of the work that has been done, including that in psychiatry, is narrowly focused on strategies for mitigating placebo effects and the challenges they might pose. In this Review, paper 2 of 2 on reconceptualising placebo and nocebo effects, we shift our focus from clinical practice to clinical trials. First, we review established and emerging approaches for navigating placebo effects in randomised controlled trials. Second, we examine three major challenges in contemporary psychiatric research that have recently highlighted the topic of placebo effects: (1) blinding and expectancy in psychedelic trials, (2) large placebo responses in interventional psychiatry trials (ie, device or procedure-based therapeutics), and (3) implications of overlapping neurobiological mechanisms between placebo effects and psychiatric treatments.

Introduction

Placebo and nocebo effects are complex neuro-behavioural phenomena mediated by expectations, beliefs, experiences, emotions, cognitive schema, and other factors related to therapeutic context.^{1–3} Placebo effects improve symptoms, whereas nocebo effects induce or worsen symptoms. Paper 1 of 2 on reconceptualising placebo and nocebo effects⁴ discusses the evolving terminology, neuropsychological models, neurobiological mechanisms, and implications for clinical practice of placebo and nocebo effects. In this Review, we focus on measuring, interpreting, and appraising placebo and nocebo effects in clinical trials. First, we provide an update on clinical trial designs that aim to mitigate the impact of placebo effects. Second, we highlight contemporary issues, such as blinding in psychedelic and behavioural interventions, navigating large placebo responses in interventional psychiatry trials, and evaluating the potential impact of shared therapeutic mechanisms between placebo effects and psychiatric treatments.

In placebo-controlled trials, the overall change observed in the placebo group (eg, decrease in a depression scale score from baseline to endpoint) constitutes the placebo response. The placebo group is used as a control to separate specific from non-specific treatment effects—ie, the placebo group is meant to experience everything that the treatment group experiences except for the active component of the treatment itself.^{1–3} This technique for determining whether a given treatment is efficacious has a long history stemming back centuries (figure 1). The first placebo-controlled trial is often credited to the Franklin–Lavoisier French Commission who designed the experiments that ultimately discredited Mesmerism in 1784.⁵ Nowadays, placebo-controlled trials remain a pillar of evidence-based medicine; however, across the research continuum, different trial phases have different goals. According to WHO, nearly half of all trials with a known development phase are placebo-controlled trials

in phase 2a, 2b, or 3a.¹³ These trials occur after safety, tolerability, and proof-of-concept studies in phase 1 and before active comparator studies in phase 3b or real-world effectiveness and optimisation in phase 4.¹⁴

The placebo response observed in the placebo group of a trial includes not only placebo effects, but also non-specific effects, comprising psychological effects and statistical artifacts. These non-specific effects include spontaneous improvement over time (rates can vary depending on variables such as the condition and disease severity), regression to the mean, the Hawthorne effect (ie, effect of being observed in a clinical trial setting), behavioural changes (eg, attending appointments to participate in a clinical trial setting), or attention and support from study personnel.¹⁵ Separating placebo effects from other non-specific effects is challenging and generally requires at least one additional trial group such as a no-treatment group or a wait-list control group. However, these trial designs are rarely used because they dilute statistical power, increase cost, and might discourage enrolment.

Whereas placebo effects are typically associated with positive changes in outcome measures, nocebo effects are typically associated with negative side effects. As described in paper 1, we note that nocebo effects are more than just the presence of negative side-effects: they might blunt or obscure therapeutic response because of negative expectations, functional unblinding, and other factors. For example, trial participants in a no-treatment group, a poorly blinded placebo group, or an active comparator group might report limited improvement or even worsening symptoms, if they suspect that the desired experimental treatment is not being initiated. Syntheses of antidepressant trials have shown that having increased likelihood of receiving placebo versus active treatment (ie, based on the number of active groups or placebo groups) is associated with a lower response rate to both placebo and antidepressant.¹⁶ Although the primary causes of nocebo effects and disappointment in these

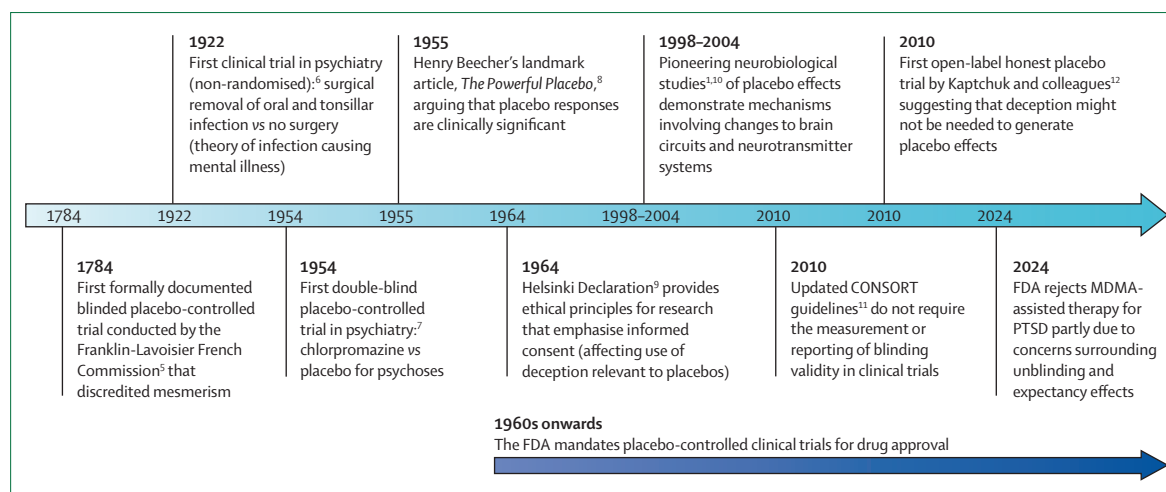


Figure 1: Historical milestones relevant to placebo effects in psychiatric research

FDA=Food and Drug Administration. MDMA=methylenedioxymethamphetamine. PTSD=post-traumatic stress disorder.

Panel 1: Impact of trial outcome measures on placebo responses

Outcomes: clinician-rated or self-reported

Placebo responses have been found to be higher for clinician-rated versus self-reported measures in individuals with panic disorder and obsessive-compulsive disorder.^{26,45} These empirical findings differ from conventional thinking that self-report measures are more prone to expectancy bias.⁴⁶ One reason clinician-rated and self-reported measures differ in placebo response rates might be that the former overestimates response through interpersonal and expectancy effects such as rater optimism or therapeutic alliance, whereas the latter could underestimate response through situational variability and the absence of social cues or reinforcement during the assessment (sometimes referred to as regression dilution bias).^{47–49} The act of the assessment necessary for clinician ratings could increase demand characteristics and Hawthorne effects. Another contextual element is the environment in which clinical outcomes are reported: familiar or private environments might reduce social inhibition and promote transparency, whereas unfamiliar or less private environments might exacerbate social anxiety and diminish disclosure.⁵⁰

Differences in scale types, range, and versions

Rating scale attributes might affect placebo response magnitudes. For example, the Hamilton Depression Rating Scale (HAM-D) and the Montgomery-Asberg Depression

Rating Scale (MADRS)—both validated, physician-administered scales for depression—broadly correlate with one another, but capture slightly different elements of the disorder. MADRS often shows slightly higher drug vs placebo differences than HAM-D, but the difference might not be clinically meaningful.^{32,51,52}

Objectively derived outcome measures

Objectively derived outcome measures might be less placebo responsive than subjective or symptom-based measures.⁸ These measures include laboratory assessments such as psychophysiological measures (eg, heart-rate variability, skin conductance, or fear-potentiated startle) and approach-avoidance tasks that have most commonly been studied in post-traumatic stress disorder, anxiety, and depression.^{53–55} New digital phenotyping approaches can leverage passive sensing and smartphone-based technologies to continuously measure real-world behaviour and physiology, including metrics such as physical activity, sleep patterns, social media usage, typing speed, voice and speech features, and eye-tracking.⁵⁶ Validation of these novel objective measures could complement laboratory-based measures in addressing placebo-response elements of clinical trials for psychiatric illness.

contexts remain incompletely understood, there are several ways to start empirically disentangling considerations such as allocation-related nocebo effects from non-response nocebo effects. First, compare outcomes between different control conditions across studies, such as no-treatment groups and wait-list groups.¹⁷ Second, use study designs that conceal the timing of active treatment or manipulate expectancy where possible and ethically appropriate.¹⁸ Third, systematically assess expectancy and

integrity of blinding. Both metrics influence clinical outcomes,¹⁹ and studies with poor blinding often show larger treatment effects.²⁰ These and other strategies will help clarify how nocebo effects affect efficacy measurements in clinical trials.

In this Review, we use the term placebo response when discussing clinical trial results. This term encompasses the net impact of interacting and overlapping positive (ie, placebo) and negative (ie, nocebo) effects, as well as

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Panel 2: Blinding and blinding integrity

Blinding or masking refers to the process by which participants, investigators, and staff are kept unaware of group assignment after randomisation. Allocation concealment prevents advance knowledge of group assignment before randomisation to reduce selection bias. Blinding integrity is crucial for disambiguating specific treatment effects from placebo responses. Without it, placebo responses can become unevenly distributed,⁵⁸ which might exaggerate effect sizes or type I errors.

Participants and study staff can often infer group assignment based on experience, contextual cues, and other factors. Discussion of expected side-effects during the informed consent process might ultimately unmask participants who experience them later in the study. It is crucial to distinguish formal (ie, blinding procedures were followed) from effective (ie, group assignment was unknown to participants) blinding.

CONSORT guidelines have focused more on reporting formal blinding than effective blinding.¹¹ Despite calls for change,⁵⁹ the latest CONSORT guidelines do not mandate empirical validation of blinding integrity.⁶⁰ Only 2–7% of trials provide such information,^{61–64} with industry-sponsored trials less likely to report on blind integrity than non-industry sponsored trials.^{62,65}

The blinding paradox

It is important to distinguish unblinding related to contextual clues or side-effects (ie, malicious unblinding) from unblinding related to noticeable clinical improvement (ie, benign unblinding). Whereas the former is an undesirable confounder, the latter is a desirable consequence but yields a paradox in which more potent treatments can be more prone to unblinding.^{66–68}

Measuring blinding integrity

The most straightforward way is to ask participants to guess their treatment allocation at the end of the treatment and to report their confidence in the guess. In a completely blinded trial, treatment guess or perceived treatment should match the actual treatment about 50% of the time in a two-arm trial—ie, participants would guess at a rate no better than chance.

Bang's blinding index is a more sophisticated variation of this method that allows participants to respond with "I do not know".⁶⁹ A new blinding quality assessment tool compatible with Bang's blinding index attempts to assess the source of unblinding to determine whether unblinding is benign or malicious, and to address additional interpretation issues.^{58,70,71} Asking for treatment allocation guesses before treatment effects are experienced might help to separate malicious from benign unblinding.

the non-specific effects previously described. We distinguish functional unblinding, which occurs when participants or study staff correctly infer group assignment despite blinding procedures, from formal unblinding, which occurs when participants are informed about their group assignment by design.

Relevance of placebo responses across psychiatric disorders

The broad use of placebo groups in psychiatric clinical trials has revealed different placebo response rates for different psychiatric illnesses. In a systematic review and meta-analysis, Bschor and colleagues²¹ compared placebo responses across nine psychiatric disorders in 9985 participants from 90 randomised controlled trials (RCTs). Participants with major depressive disorder (MDD) showed the most improvement, followed by those with generalised anxiety disorder (GAD). Panic disorder, ADHD, post-traumatic stress disorder (PTSD), social phobia, and mania followed in descending order, with obsessive-compulsive disorder (OCD) and schizophrenia showing the least improvement (appendix p 1). In meta-regression analyses, only gender showed a statistically significant association—women showed more improvement in response to placebo than men. Another synthesis of this topic, by Huneke and colleagues,²² found similar trends analysing data from 20 meta-analyses, summarising 1691 RCTs involving 261730 participants (appendix p 1).

Together, these meta-analyses highlight the importance of continuing to examine placebo responses across psychiatric disorders, and particularly for mood and anxiety disorders. Placebo responsiveness is not uniform, and this variability has implications for both trial design and data interpretation. These transdiagnostic comparisons summarise placebo responses across decades of trials, but they do not capture the magnitude of change within each diagnosis over time. There is emerging evidence that placebo response is dynamic rather than static. For example, OCD and schizophrenia show notable increases in the magnitude of placebo responses over time. Systematic reviews and meta-analyses have captured these temporal trends, highlighting how within-diagnosis factors could shift the relative placebo responses across diagnoses with changes in patient factors, diagnostic trends, and trial designs.^{23–27}

Although the results derived from these meta-analyses of placebo responses across psychiatric conditions highlight the importance of continuing to use placebo controls in RCTs, such use of placebo raises ethical questions. For example: under what circumstances is it ethical to withhold an active treatment? Such questions are outside the scope of this Review but have been discussed in detail elsewhere.^{28,29} The following sections examine both traditional and contemporary issues of placebo responses in psychiatric research with relevance across psychiatric disorders.

See Online for appendix

Established and emerging strategies for navigating placebo effects

Developing treatments for psychiatric and pain disorders can be challenging because of placebo response rates upwards of 20–40%.¹⁵ For example, SSRIs are the most prescribed medications for one of the most common psychiatric disorders (ie, MDD). A comprehensive meta-analysis reported a statistically significant difference between SSRIs and placebo.³⁰ However, the magnitude of difference was only 2 points on the Hamilton Depression Rating Scale (HAMD)-17 scale. This magnitude of change corresponds to 0.3 standardised mean difference, which is typically regarded as a small effect size. A patient level meta-analysis corroborated that 80% of the benefit of antidepressants could be attributable to placebo response; however, the authors found a trimodal response distribution. Based on data from 73 388 patients, about 15% of patients experience substantial antidepressant effect beyond placebo.³¹ Statistical significance is not equivalent to clinical significance and there are ongoing debates regarding the best way to synthesise and interpret these data.³²

There is also disagreement about whether placebo responses are increasing over time. One study reported an increase of 7% per decade,³³ but these findings have been challenged.³⁴ A meta-analysis of neuropathic pain studies reported that placebo responses were increasing in the US, but not worldwide.³⁵ Recognising the difficulty of overcoming placebo effects, trialists have developed several strategies that aim to mitigate placebo effects. Mitigating placebo effects is often called derisking because the goal is to reduce the probability of a non-significant difference between active treatment and placebo. We discuss two clinical trial design models that aim to accomplish this goal.

Placebo run-in is a design wherein the main trial (ie, phase 2) is preceded by an initial treatment phase (ie, phase 1) in which all participants receive placebo. With this design, patients whose symptoms respond to placebo in phase 1 are excluded from phase 2 randomisation, and phase 1 data are excluded from efficacy analyses. Placebo run-in designs vary in terms of study phase lengths and what level of detail about the trial design is shared with participants, but the common goal is to reduce responses to placebo in phase 2. However, it is unclear if this study design achieves this goal. A recent systematic review of trials of antidepressants found that the likelihood of response in trials with run-in periods (odds ratio [OR] 1.89, 95% CI 1.76–2.03) was not larger than in trials without run-in periods (1.77, 1.65–1.89; $p=0.18$), nor was the between-group effect size larger in trials with run-in periods (Hedges' $g=0.33$, 95% CI 0.29–0.38) than without run-periods (0.34, 0.30–0.38; $p=0.92$).³⁶

A more sophisticated version of the placebo run-in is the sequential parallel comparison design (SPCD),³⁷ in which both phases involve randomisation to active or placebo

treatment. In phase 1, participants are randomly assigned in an unbalanced manner so that more participants receive placebo than active treatment. In phase 2, participants whose symptoms do not respond to placebo are randomly assigned once more and participants whose symptoms responded to placebo are excluded. Those who were randomly assigned to active treatment in phase 1 do not participate in phase 2, but their data are used in the efficacy analysis. Similar to the placebo run-in design, SPCD aims to reduce placebo response in the efficacy analysis. The advantage of SPCD over placebo run-in is that data from both stages are utilised to assess efficacy. Therefore, SPCD has more power for a given sample size under most circumstances.³⁸ There is no systematic review of SPCD trials, but there are some data to suggest that SPCD increases the probability of positive trial results.³⁹

Study designs that aim to mitigate placebo effects have some drawbacks. First, mitigating placebo effects could introduce selection bias—ie, discarding data from people whose symptoms respond to placebo means that the analysed sample is no longer a random sampling of the target population. Therefore, trials that mitigate placebo effects have less external validity and less generalisable results.⁴⁰ However, generalisability might not be a primary concern for some trial phases (eg, proof-of-concept studies). Second, mitigating placebo effects could increase the chances of a participant receiving placebo or increase the length of time that a participant waits before receiving active treatment. Third, mitigating placebo effects does not necessarily develop more effective treatments. Instead, this approach might reduce the standards against which a new treatment is measured by excluding those whose symptoms respond to placebo.

Recent developments in artificial intelligence and machine learning could offer an alternative approach. For example, Cognivia have developed a psychological profiling questionnaire that is intended to be administered to study participants at baseline. Machine learning is applied to data from the questionnaire and other baseline data to estimate a participant's placebo responsiveness. This score could theoretically be used as a covariate in efficacy analyses. Such adjustments with composite covariates have recently been accepted in a US Food and Drug Administration (FDA) guidance⁴¹ dedicated to the use of baseline covariates to adjust statistical analysis. However, the accuracy and potential utility of this type of placebo score in predicting placebo response is still under investigation. In an RCT of 180 individuals with osteoarthritis, the Pearson correlation between the predicted and the true placebo response was 0.46–0.60 across outcomes, corresponding to an R^2 range of 0.21–0.36⁴² (ie, the novel placebo score covariate explained 21–36% of the variance). Adding this covariate increased the precision of the estimated treatment effect by 40%, facilitating the conclusion that placebo had a role in the small between-group difference.

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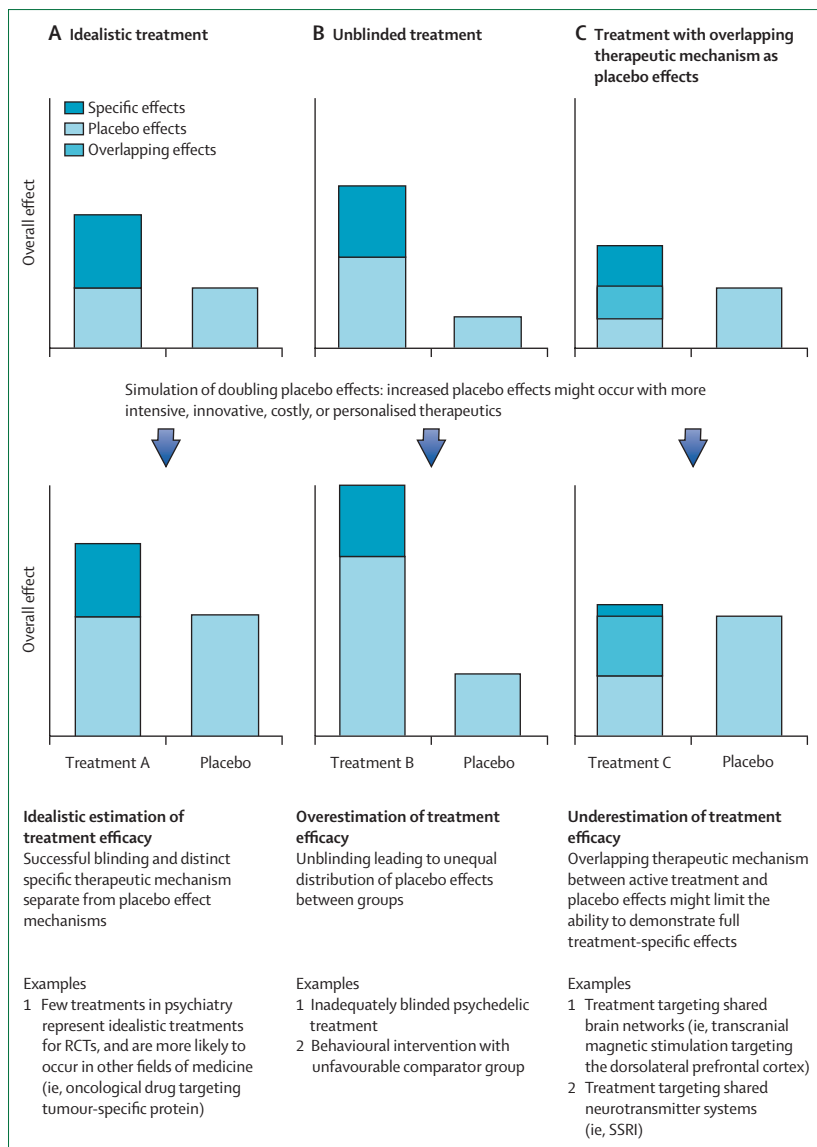


Figure 2: Modelling how factors related to placebo effects can affect measurements of clinical efficacy

Hypothetical treatment trials (A–C) aim to provide a theoretical model for how factors related to placebo effects can affect the interpretation of clinical trial results. Treatment-specific effects are kept constant across the three hypothetical treatments and other non-specific effects that might contribute to the overall placebo response are not depicted. (A) Placebo effects are split evenly between groups (ie, patients are adequately blinded and do not know what treatment they are receiving). When placebo effects are elevated, they are increased equally in both groups, preserving the ability to detect a between-group difference. (B) Unblinding might lead to unequal distribution of placebo effects, resulting in potential overestimation of treatment efficacy (ie, patients in the treatment group having higher expectations as they believe they are receiving treatment and patients in the placebo group having lower expectations as they believe they are receiving placebo). This effect of unblinding was hypothetically modelled as lowering placebo effects by half in the placebo group and increasing by half in the active group. (C) Treatments with overlapping therapeutic mechanisms with placebo effects could have underestimation of their specific efficacy, particularly if placebo effects are large (ie, ceiling on shared effects). If such treatments have large specific effects, they could still be found to be more efficacious than placebo (overcome any potential depicted underestimation). Primary limitations of modelling: (1) the impacts related to placebo effects shown in B and C are not mutually exclusive (ie, there could be treatments that have both blinding challenges and overlapping mechanisms), making interpretation more complex than these sample scenarios in isolation; (2) modelling does not consider potential synergistic interactions between specific and placebo effects; and (3) other non-specific effects are not depicted and for some examples (eg, an inadequately blinded psychedelic treatment), it remains unclear how much of the potential unequal distribution modelled might be driven by placebo effects versus other non-specific effects. RCT=randomised controlled trial.

Finally, the context in which treatment outcomes are measured also affects placebo response magnitude.^{43,44} In panel 1, we provide an overview of relevant outcome measure characteristics, including clinician-rated, self-reported, and psychophysiological approaches. New strategies, such as the latter objectively-derived markers, might temper placebo responses, but do not help to reconcile the clinical dilemma in which objective and subjective measures are not aligned.

Blinding and expectancy in psychedelic research

Blinding is crucial for ensuring an even distribution of placebo responses (ie, placebo effects or other non-specific effects as described previously) between trial groups and thereby allowing effects above placebo to be attributable to specific treatment effects. If blinding is inadequate, stronger placebo effects might be observed in the active treatment group and weaker placebo effects in the placebo group due to divergent expectancies surrounding receiving the desired treatment.⁵⁷ Figure 2A shows an idealised trial where the placebo control condition is working as intended, whereas Figure 2B depicts the potential unequal distribution of placebo effects between groups and consequent inflation of the measured treatment response (ie, expectancy bias). In panel 2, we provide a primer on blinding methods, integrity, and standards in clinical research.

Psychedelic research has emerged as one of the most exciting new frontiers in psychiatry. However, blinding difficulties and expectancy in psychedelic research have caused placebo effects to receive more attention in mainstream medical discourse than ever before. The acute psychoactive effects of psychedelics present unique challenges to blinding and has led to trial results being called into question. In June, 2024, the FDA Advisory Committee held a hearing to review methylenedioxymethamphetamine (MDMA)-assisted therapy for PTSD—only one of two phase 3 trials assessed blinding. In that trial, 82 (85%) of 96 participants correctly guessed their treatment allocation of MDMA versus inert placebo.⁷² The FDA declined to approve MDMA citing various methodological concerns, although the agency's reasoning did not exclusively focus on functional unblinding.

In most psychedelic-assisted psychotherapy trials, the correct guess rate is approximately 90% (appendix p 3).^{73–79} Some of these trials are still described as double-blinded, despite the authors not demonstrating effective blinding. To date, the most effective blinding in a classic psychedelic trial has a correct guess rate of 79%,⁸⁰ far from the 50% that represents random chance guessing. To assess unblinding, correct guesses related to side-effects and contextual factors (malicious unblinding) should be separated from correct guesses related to noticeable improvement of the treated condition (benign unblinding).⁶⁶ Malicious unblinding is problematic

because it means that trial results could be false positives caused by unblinding bias. By contrast, benign unblinding is desirable because it means that treatment had substantial benefits. This interpretational issue has been used as an argument against assessing blinding integrity.⁶⁷ To address this problem, treatment guesses could be collected before the delayed benefits of treatment or with additional questions about the reason for the guess.⁷⁰

Previous research has highlighted that blinding could also be compromised in trials of non-psychedelic medications, such as SSRIs. However, the evidence on this claim is mixed, with two recent analyses having reached opposite conclusions.^{65,81} These differing conclusions can be attributed to methodological differences. Nonetheless, the authors of both analyses agree that the correct guess rate in modern antidepressant trials is approximately 60% (appendix p 3), which is considerably lower than the approximate 90% commonly observed in psychedelic studies. Even if blinding is compromised in traditional antidepressant trials (depending on the statistical model used), the magnitude of unblinding is much smaller and hence so is the magnitude of the associated biases.

To date, only one study of psilocybin for the treatment of depression has assessed expectancy, which would provide insight into how conscious expectations and other potential unconscious processes could be affecting outcomes.⁵⁸ The trial was a head-to-head comparison of escitalopram and psilocybin for treatment-resistant depression.⁸² Baseline expectancy was associated with outcomes in the escitalopram group, but not the psilocybin group. Statistical modelling suggests that if expectations in the escitalopram group had been higher, then the between-group difference would have been eliminated.^{71,83}

A network meta-analysis found that the within-group (ie, the change from baseline to endpoint) change in the

placebo condition was much lower in psychedelic studies than in trials of traditional antidepressants.⁸⁴ In psychedelic trials, the response in the placebo group was 3.8 points less on the HAM-D-17 scale compared with the response in the traditional antidepressant trials. This difference is about twice the difference between placebo and SSRIs. Aligned with this result, the interim data analysis of a phase 2 trial of psilocybin for the treatment of depression had the very unusual finding that participants in the placebo group had worsening depression.⁸⁵ In a meta-analysis of modern antidepressant trials, all 304 placebo groups that were analysed showed improvement.³⁰ A plausible explanation for the absent or reduced placebo responses in psychedelic trials might be nocebo effects: unblinding and disappointment (lower expectations of improvement or even negative expectations) when participants realise they are receiving the control condition.

Most psychedelic trials have strict inclusion and exclusion criteria to mitigate the risk of complications. Due to these strict criteria, the generalisability and external validity of these trials have been questioned.⁸⁶ With psychedelic services becoming more legally available, and hence more real-world evidence becoming available, the potential differences between efficacy versus effectiveness are of interest.

Strategies for improving blinding in psychedelic trials

There is no easy or perfect solution to blinding challenges facing the psychedelic field, but several approaches have emerged that might improve blinding integrity and help separate expectancy from drug effects (table). One commonly used approach is to use active placebos, which are intended to mimic side-effects of the active treatment without a specific therapeutic mechanism. So far, this strategy has had little success with psychedelic treatments. A study using

	Description	Advantages and disadvantages
Active placebo	Participants in the placebo condition receive an active drug that mimics the side-effects of the active treatment.	Difficult to find a suitable active placebo for most drugs. Ideally, the active placebo should have all the non-specific effects of the active treatment, but none of the specific effects—ie, it should mimic all the side-effects, but none of the therapeutic effects.
Incomplete disclosure of trial design	Participants receive only partial information about the trial design, which can help to manage their expectations.	Easy to implement and routinely used in medical trials. Trade-off between providing participants with all the necessary information for informed consent and hiding trial design. Blinding effectiveness likely depends on exactly what information is shared or hidden.
Selective patient population	Participants recruited are psychedelic-naïve or without positive expectations for psychedelics.	Participants who are psychedelic-naïve might not recognise treatment, hence the improvement of blinding integrity. Even if blinding is not successful, expectancy bias should be minimised if patients have lower positive expectations about psychedelics. Overall, this method has increased difficulty of recruitment.
Balanced placebo designs	A 2 × 2 design, where one factor is the received drug (active or placebo) and the other factor is what patients are told about the drug (told active or told placebo). This method can be incorporated into both parallel-groups and crossover designs.	Can differentiate true drug vs true placebo effect (ie, received-active, told-placebo vs received -placebo, told-active) and can test drug-placebo interaction effects. Participants might not believe what they are told about the drug condition. Ethical and legal dilemmas about deception might be mitigated by authorised deception. Design might induce negative psychological effects in participants who were told they have received the placebo but received the active drug (or vice versa).
Open or hidden paradigm	All participants receive active treatment. Drug administration mimics normal clinical care for the open group. For the hidden group, the drug is administered by a timer controlled pump without the typical therapeutic context (eg, presence of doctors).	Can measure placebo component of treatment—ie, the difference between open and hidden administrations, even without a placebo group. Ethical concerns about hidden administration.

Table: Trial designs to improve blinding in psychedelic research

diphenhydramine versus psilocybin reported a 94% accurate guess rate and a study using niacin versus psilocybin reported a 100% accurate guess rate.^{77,87} One study developed a ayahuasca placebo that tasted similar to ayahuasca and induced nausea.⁷⁴ This active placebo successfully masked five (33%) of 15 participants in the placebo group; however, all 14 participants in the ayahuasca group correctly identified their treatment allocation.

Another strategy is to use very low doses of the medication being studied as an impure placebo, based on the assumption that the lowest dose causes mild psychoactive effects without therapeutic effects. The largest psilocybin trial to date used this dose–response design, comparing a single dose of 1 mg, 10 mg, and 25 mg in treatment-resistant depression.⁸⁸ In this study, the 1 mg dose served as a low-dose placebo. However,

blinding integrity data were not collected. At the 3-week primary endpoint, 10 mg outperformed 1 mg, and 25 mg outperformed 10 mg, suggesting a dose–response relationship. However, functional unblinding could be greater in the higher dose group, complicating the interpretation of results. Lastly, there are more unconventional methods, such as the balanced-placebo design or open-hidden design. These designs interrogate placebo responses by manipulating participant expectations about assignment. For example, a participant might be told that they are receiving a placebo when they are receiving an active treatment, or vice versa.⁸⁹ These are intriguing strategies worthy of further study, but such methods have many limitations, including ethical concerns surrounding deception.⁸²

There are emerging analytical approaches that could be used to try to correct for issues of unblinding and

Panel 3: Strategies for managing blinding and placebo effects in behavioural interventions

Comparison with active therapies

For comparative effectiveness designs in which an experimental intervention is compared to a structurally equivalent active therapy (eg, a different psychotherapy modality), it might be possible to accomplish informed consent without revealing the unique content of either intervention. This approach requires a balance of providing sufficient details about what participation in the study involves and potential harms with intentional vagueness about the distinguishing features of each group. The consent form description could focus on the common structural aspects of the intervention (eg, details about who the participant will meet with, how often, and for how long).

Comparison with structurally non-equivalent active interventions

For comparison designs in which the intervention is not structurally equivalent (eg, comparing delivery methods, doses, or a full treatment package to select treatment components), pre-consent randomisation (Zelen design)⁹⁴ or a two-staged consent process can preserve blinding to the non-assigned group.⁹⁵ In a two-stage process, participants first consent to assessments and concealment, then after randomisation they are asked to consent to the intervention to which they are assigned. Details about the non-assigned intervention are withheld until study completion. In our experience, it is rare (<5%) for participants to consent at stage one and decline at stage two.⁹⁶

Comparison with placebo-like therapy

As a placebo-like control intervention becomes more credible, it generally becomes less inert,⁹⁷ resulting in underestimation of the treatment effect. Non-directive counselling is a widely used placebo-like control condition that is not inert,⁹⁸ delivering expectations for improvement and psychotherapy common factors (eg, empathic listening), which might account for considerable variance in treatment outcomes.⁹⁹ Whichever placebo condition is selected, the investigators should attempt

to increase its credibility, such as by making it structurally equivalent¹⁰⁰ and providing a rationale for how the intervention is logically related to the participant's concerns.⁹⁷ Unlike drug and device studies, non-specific effects (including attention and expectancy effects) can be considered essential components of a behavioural intervention, not a competing explanation for its efficacy.¹⁰¹

Comparison with no study care

Control groups that involve no researcher-provided intervention (eg, usual care and waitlist conditions) might be appropriate when there are no established therapies for a condition. Mitigating bias is still important in such designs. Participants in a waiting list group typically experience less reduction in their symptoms than those enrolled in an observational study,¹⁰² perhaps because the open-label assignment modifies expectations and behaviour.¹⁰³ A deception waiting list (unknownst to the participant) might be ethical and feasible when the waiting list does not grossly exceed typical waiting times for clinical services. This strategy should mitigate the nocebo effects associated with being told to wait, but to our knowledge, has not been empirically evaluated.

Additional strategies for any trial design

Blinding outcome assessors to participant allocation is typically offered as a solution to being unable to mask participants,¹⁰⁴ but many behavioural intervention studies use self-reported primary outcome measures, such that the outcome assessor is the participant.¹⁰⁵ Unmasked study participants might unmask raters unintentionally by referring to their treatment during the rating interview. When bias cannot be mitigated, it should at least be measured. The CONSORT extension for social and psychological interventions recommends that if masking was not done (eg, because it was not possible), authors should describe the methods, if any, used to assess performance and expectancy biases.¹⁰⁴

unbalanced distribution of placebo effects between groups. One method, called the Correct Guess Rate Curve, is a computational pipeline that estimates what would be the outcome of a perfectly blinded trial based on data from an imperfectly blinded trial.⁷⁰ The method works by splitting the data into four strata according to treatment and guess, and then resampling these strata such that the obtained sample has a correct guess rate of 0·5—ie, a perfectly blinded two-armed trial. Another strategy is to use baseline expectancy scores as covariates to adjust the trial results.^{58,71,83} This method does not correct for all factors associated with unblinding, but potentially controls for expectancy bias, which is believed to be a major component of the wider unblinding bias.

The techniques previously discussed focus on participant blinding. Functional unblinding can also affect investigators, therapists, and raters. A common problem is that raters might inflate baseline scores to enrol new participants, which could lead to a biased estimate of the treatment effect. This rater-blinding issue can be eliminated using third-party raters, who have no knowledge about the trial design or the timepoints at which participants are assessed. For example, a phase 3 MDMA trial used a central pool of blinded independent assessors.⁷²

In summary, it remains unclear whether existing strategies can improve blinding in psychedelic studies. The inability to conduct a truly blind placebo-controlled psychedelic trial does not mean they are not effective. Growing evidence shows cellular-level and circuit-level neurobiological mechanisms supporting the credibility of meaningful psychedelic-specific effects.⁹⁰ Although the gold standard efficacy marker of double-blinded RCTs should be respected, a truly blinded psychedelic trial might be unattainable, and some researchers suggest that this limitation should just be accepted. This line of reasoning remains an area of contentious debate, but proponents would emphasise that the end goal of medicine is to help people in real life settings, not to demonstrate efficacy in idealised trials.^{92,93}

Behavioural intervention trials

There is ongoing debate about the appropriateness of placebo-like controls in behavioural intervention studies. Blinding participants to the treatment they are receiving and providers to the treatment they are delivering is challenging with behavioural interventions that require active participation from both parties (eg, psychotherapy, exercise, and mindfulness). This challenge should not absolve investigators from the duty to attempt to mitigate bias related to potential unblinding. Expectation of benefit may be driven not only by the characteristics of the assigned intervention, but also by perceptions of its potency relative to the alternatives—knowing what other treatments one could have been assigned, and which is hypothesised to be superior could contribute to

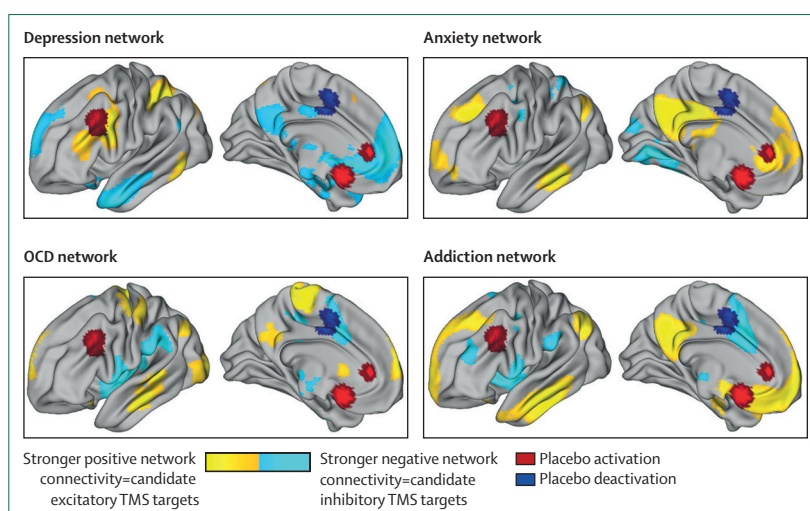


Figure 3: Overlap of placebo neuroimaging correlates with symptom-network psychiatric treatment targets Placebo activations (dark red) and deactivations (dark blue) overlap with symptom-network targets for four US Food and Drug Administration-cleared TMS indications: depression, anxiety (when comorbid with depression), OCD, and addiction. Depicted network targets were generated using causal network mapping which involved: (1) mapping functional connectivity patterns of lesions (addiction network), brain stimulation (OCD) sites or their combination (depression and anxiety networks) using the human connectome, and (2) correlating lesion or brain stimulation site connectivity patterns with clinical data to identify voxels associated with symptom change. Depicted networks were Z-scored and thresholded at $|Z| > 1.5$. Positive network correlations (labelled in yellows) reflect candidate excitatory TMS targets; negative network correlations (labelled in blues) reflect candidate inhibitory TMS targets. Spatial overlap is shown descriptively. OCD=obsessive-compulsive disorder. TMS=transcranial magnetic stimulation.

performance bias. Partial blinding strategies can be used to withhold this information from participants and providers. The choice of strategies is constrained by the comparator group best suited to the research question (panel 3).¹⁰²

Surgical, device or procedure-based therapeutics and interventional psychiatry

Over the past decade, there has been a surge in the development of surgical, procedural, and device-based interventions in psychiatry. Interventional psychiatry treatments are challenging to study because of potential increased placebo responses and complex ethical considerations governing randomised controlled trials (RCTs).^{106,107}

Compared with oral medication, these treatments are typically more intensive and elaborate, have higher perceived innovation, higher cost, and receive more media attention—all factors associated with increasing placebo effect magnitudes.^{108–110} Additionally, these treatments are often provided with more extensive, in-person care in specialised settings such as hospital-based treatment centres or tertiary care centres, which could increase positive expectancies related to the therapeutic context. A simulation of increasing placebo effects across hypothetical treatment scenarios is modelled in figure 3.

Challenges navigating large placebo responses related to surgical, procedural, or device-based interventions are not unique to psychiatry and have a long history in medicine

Panel 4: Challenges in interventional psychiatry

Interventional psychiatry treatments evolve over time

Procedural treatments are generally more dynamic than medications with established pharmacokinetics, pharmacodynamics, and dose–response curves. New treatment parameters, targeting, dosing strategies, and other factors complicate the process of determining when a procedure is established and therefore when equipoise is lost between active and sham procedures. Electroconvulsive therapy is considered one of the most effective treatments in psychiatry,¹²² however, there are still criticisms that sham electroconvulsive therapy has been used too sparingly in trials. Existing sham-controlled electroconvulsive therapy trials show separation between active and sham, but concerns related to design limitations and possible unblinding have been raised.^{122,127} Trials testing newer forms of convulsive therapy, such as magnetic seizure therapy, often use active comparators rather than sham interventions.¹²⁸ Active comparator designs deliver active treatment to all participants and can bias investigators against finding a between-group difference, but they provide few insights into the magnitude of the placebo response in either group.¹²⁷ Similar concerns apply to treatments such as neuroimage-guided accelerated transcranial magnetic stimulation (TMS), for which an RCT with intact blinding had to be discontinued because of clear separation between active and sham interventions.¹²⁹ The US Food and Drug Administration cleared the device tested in this trial. Now, there are questions as to whether there is clinical equipoise between sham and real accelerated TMS, posing challenges for confirmatory sham-controlled efficacy trials with larger sample sizes across multiple sites.

Interventional psychiatry therapeutic context evolves over time

Individuals currently receiving electroconvulsive therapy have illnesses with longer durations, greater severity, and more complex comorbidity profiles than those receiving the therapy in previous decades.^{130,131} Many of these individuals have already tried treatments such as TMS and esketamine before they consider electroconvulsive therapy. The wide accessibility of online information and social media might influence treatment expectations and desirability in unpredictable ways. These variables likely influence how an intervention performs in different clinical trials at different times, even when the intervention itself has not changed. For example, a treatment that becomes less novel and less desirable over time might result in placebo effects becoming nocebo effects. A high-profile comparative effectiveness trial in interventional psychiatry—an open-label, randomised, non-inferiority trial of intravenous ketamine versus electroconvulsive therapy for treatment-resistant major depressive disorder (MDD) without psychosis—exemplifies this issue.¹³² The primary finding of non-inferiority could have been driven by a remarkably low response rate for open-label electroconvulsive therapy. All trial

participants had already been prescribed electroconvulsive therapy, a stigmatised, century-old intervention covered by insurance. Ketamine, a recently developed antidepressant intervention, would be perceived as more novel, advanced, expensive, and inaccessible except through study participation. The incentive to enrol in the study was the opportunity to be randomly assigned to ketamine. Disparate dropout rates after randomisation but before treatment initiation (four [2%] of 200 participants in the ketamine group vs 31 [15%] of 203 in the electroconvulsive therapy group) support concerns about sampling bias, expectancy bias, self-selection bias, and other confounders.

Vulnerable patient populations

Interventional psychiatry treatments are typically prescribed for individuals with severe or treatment-resistant psychiatric illnesses, whose symptoms are generally considered less responsive to placebo.¹³³ These patients are often considered vulnerable populations, where additional precautions and rationale might be needed to justify potential randomisation to placebo. One MDD trial of theta-burst TMS found very large placebo responses in patients with low refractoriness but not in patients with moderate-to-high refractoriness.¹³⁴ However, a recent meta-analysis of placebo responses in treatment-resistant depression trials still reported a large overall effect size (Hedge's g 1.05, 95% CI 0.9–1.1).¹³⁵

Blinding challenges

Devices and procedures in interventional psychiatry pose greater blinding challenges than placebo pills. Early TMS studies probably were affected by unblinding due to failure of mimicking the auditory, tactile, and visual components of active treatment (eg, tilting the coil away from the head). The development of specifically engineered sham systems offer substantial improvements,¹³⁶ and accordingly some syntheses have reported increased placebo responses in TMS over time.¹³⁷ This offers a prime example of how the quality of blinding can affect trial results.

In trials of ketamine and esketamine, midazolam or saline are often used as the sham intervention but neither mimics the signature dissociative elements of ketamine. Midazolam might improve blinding to some extent,¹³⁸ and the effect size of ketamine is smaller in trials that administered midazolam than in those that administered saline.¹³⁹ A recent randomised trial that successfully blinded intravenous ketamine treatment versus intravenous saline (as MDD patients receiving treatment were under anaesthesia for an elective surgery), showed very large placebo responses that were not significantly different from the ketamine group.¹⁴⁰ Critics argue that the dose and administration were not aligned with established protocols for depression, but nevertheless, this trial raised important questions about placebo effects.

more broadly. Trials yielding unexpectedly large placebo responses include: arthroscopic surgery for osteoarthritis of the knee,¹¹¹ percutaneous coronary intervention for stable angina,¹¹² renal denervation for treatment-resistant hypertension,¹¹³ spinal cord stimulation for chronic radicular pain,¹¹⁴ vertebroplasty for osteoporotic fractures,¹¹⁵ arthroscopic subacromial decompression for shoulder pain,¹¹⁶ and stem-cell treatments for Parkinson's Disease.¹¹⁷ These negative trials generated considerable controversy in their respective fields as they did not fail because no participants got better, but because all participants got better regardless of whether they received active or sham treatment. These trials did not drive large-scale interrogations into placebo effects; instead, they often triggered post hoc alternative explanations about how sham interventions must have some sort of active component. There are few RCTs in medicine directly comparing placebo treatments with varying intensiveness because of logistical and ethical challenges. However, one RCT found that sham devices had greater effects than placebo pills on the primary pain outcome.¹¹⁸

Procedural or device-based treatments for more severe or refractory psychiatric illnesses are often under the subspecialty of interventional psychiatry.^{119–121} Transcranial magnetic stimulation (TMS), deep brain stimulation (DBS), electroconvulsive therapy, vagal nerve stimulation, focused ultrasound, cranial electrical stimulation, and rapid-acting pharmacotherapies—such as ketamine, esketamine, and brexanolone—are among the procedural interventions that might be considered part of this framework.^{122–124} Meta-analyses have shown that active TMS, esketamine or ketamine, and electroconvulsive therapy all separate from their respective sham counterparts^{122–124} and outperform active comparators or alternative treatments in head-to-head trials.^{125,126} However, interventional psychiatric treatments have many unique challenges with regard to increased placebo effects, trial methodology, blinding integrity, and regulatory approval (panel 4).

There have been many notable examples of placebo responses that were larger than anticipated in psychiatric trials of procedural or device-based treatments, particularly in the depression literature. For example, two large, multicentre RCTs for MDD showed remission rates of 37% with sham TMS¹⁴¹ and 42% with sham cranial electrical therapy.¹⁴² A multicentre vagal nerve stimulation study in individuals with treatment-resistant depression similarly reported a sham response rate that was higher than expected, with a late surge in the sham group that resulted in no significant difference between active and sham on the primary outcome (percent time in Montgomery-Åsberg Depression Rating Scale [MADRS] response).¹⁴³ Unsurprisingly, none of these trials showed separation of active from sham. Despite these examples, it remains unclear whether more intensive treatment modalities in psychiatry have larger placebo effects. A meta-analysis of placebo responses from trials of

treatment-resistant depression reported a large pooled placebo effect size (Hedge's g 1.05, 95% CI 0.9–1.1), equating to a 30–50% reduction in depression symptoms on standard outcome measures (eg, HAMD or MADRS).¹³⁵ No significant differences were reported between placebo or sham treatment modalities (eg, sham device *vs* inert pill). However, heterogeneity in placebo responses across studies was high. For example, one TMS study reported no improvement in the sham group, whereas another reported 50% improvement in the sham group. Placebo groups are supposed to be relatively standard benchmarks for comparison. These large between-study discrepancies should raise concern for differential blinding integrity and expectancy bias across studies. Such factors could have undermined the ability to detect potential differences between modality types. There is a clear need for more systematic approaches to appraising placebo effects in clinical trials¹⁴⁴ and potential factors driving the observed variability remain incompletely understood. For example, an accelerated TMS protocol with a large effect size showed only 11.1% reduction using the MADRS in the sham group despite intact blinding. This trial was discontinued at the planned interim analysis, as there were ethical concerns about continuing to give patients with depression sham treatment when active treatment was superior.¹²⁹

Lastly, there is a growing movement in interventional psychiatry to offer personalised treatments adapted to the specific neurobiology of the patient. The novelty and innovation associated with such personalisation might add to the list of factors capable of influencing placebo responses. A 2023 study in healthy volunteers using a pain paradigm found that framing a sham treatment as personalised to their genetics and physiology was more efficacious than the same sham treatment that was framed as effective generally.¹¹⁰ Studies that compare individualised with non-individualised interventions in an effectively blinded manner are crucial for advancing the field.

Overlap in therapeutic mechanism between placebo effects and psychiatric treatments

Most studies and analyses attempt to separate the hypothesised therapeutic mechanism of an active intervention from mechanisms associated with placebo, nocebo, and other non-specific effects. However, this distinction might not be straightforward, and their potential overlap has major implications for interpreting results from clinical trials. As discussed in paper 1 on reconceptualising placebo and nocebo effects,⁴ the pathophysiology of many psychiatric disorders (and other brain-based disorders in medicine) overlaps with neurobiological systems that are modulated by placebo effects.¹ Such overlap extends to the targets and mechanisms of treatments for psychiatric disorders. The foundation for this concept was demonstrated more than 20 years ago in a seminal depression trial with PET neuroimaging of both placebo and SSRI groups.¹⁰ In this trial, Mayberg and colleagues identified a common

Search strategy and selection criteria

We searched PubMed, Google Scholar, and our personal files for articles published between Jan 1, 1950, and July 15, 2025, using the search terms “placebo”, “nocebo”, “psychiatry”, “clinical trials”, “blinding”, and “expectancy”. Articles resulting from these searches and relevant references cited in those articles were reviewed. Only articles published in English were included in our references.

pattern of regional metabolic changes in people whose symptoms responded to placebo and people whose symptoms responded to fluoxetine, which is suggestive of a shared mechanism for depression remission, regardless of treatment modality. We summarise the accumulating evidence for the overlap between placebo effect and psychiatric treatment therapeutic mechanisms in the appendix (pp 5–6). This summary includes overlap at the genetic, neurotransmitter system, and brain circuit levels. We highlight brain circuit level, as the advent of targeted neuromodulation in psychiatry has provided some of the most direct evidence that brain regions and circuits targeted by brain stimulation treatments for psychiatric disorders might overlap with brain regions and circuits implicated in placebo effects. A meta-analysis highlighted regions such as the dorsolateral prefrontal cortex (DLPFC) and subgenual anterior cingulate cortex—as well as connected regions implicated in reward and expectation—that are activated during placebo responses and are also the specific targets of neuromodulatory depression treatments (TMS and DBS).¹⁴⁵ Figure 3 shows such overlap between brain circuit targets identified with convergent causal mapping¹⁴⁶ for FDA-cleared TMS indications (depression, anxious depression, OCD, and addiction) and placebo activations and deactivations.

Potential overlapping therapeutic mechanisms between placebo effects and specific treatments have major implications for measuring clinical efficacy. For example, if placebo effects and TMS treatment for depression both increase activity of the left DLPFC (and connected circuits), and placebo effects are large in the given therapeutic context, then the DLPFC circuit could be near maximally activated through placebo effect mechanisms (ceiling effect) and TMS might have little ability to demonstrate an incremental specific effect.¹⁴⁵ The statistical implications of this theoretical model of so-called stolen effects is depicted in figure 2C. This situation may be compared with an idealistic treatment that has a specific therapeutic mechanism entirely distinct from placebo effects (eg, an antibody-based cancer treatment with a gene product target in the liver), where, even if placebo responses are large, they should be equally added to both groups without affecting the specific treatment effect, preserving the ability to detect a treatment difference.

Another model relevant to overlapping therapeutic mechanisms is one in which placebo effects and

psychiatric treatment effects might be synergistic. For example, positive expectation could activate the DLPFC circuit and stimulation of this circuit in this heightened state-dependent context could have potentiated effects.¹⁴⁷ Supportive of this concept, a naturalistic cohort study of people with treatment-resistant depression revealed that higher pre-treatment positive expectations were associated with greater remission rates with open-label repetitive TMS stimulating the left DLPFC.¹⁴⁸ However, this study was not designed to analyse synergism specifically and could just reflect placebo effect contributions to the overall therapeutic effect.

This topic links to the evolving psychedelic therapy literature. It has been a long-held belief in psychedelic culture that set and setting influence psychedelic experiences. Setting is defined as the physical, social, and cultural environment, whereas set refers to the mindset and psychological state (eg, expectations, mood, and intentions).^{25,149} Some authors have speculated that the placebo response and psychedelic drug effects might be similarly synergistic.¹⁴⁹ Further research is needed to study such theoretical models and their potential interaction empirically. Systematically collecting data on and potentially manipulating pre-treatment expectancy (ie, open-hidden paradigms or balanced placebo designs; table) would be good initial objectives for future trials.

Conclusions

There is a long history of trying to navigate and mitigate placebo responses in psychiatric research. Although these efforts have been important in advancing the field, contemporary issues surrounding placebo and nocebo effects—including challenges related to blinding and expectancy in psychedelic and behavioural interventions, placebo responses that are larger than anticipated in device or procedural psychiatric treatments, and evolving understanding of potential shared neurobiological mechanisms between placebo effects and psychiatric treatments—highlight the need for further careful consideration of placebo effects. We propose the following areas as crucial to advance this topic: (1) establish systematic methods for data collection on pre-treatment expectancy and blinding in RCTs; (2) expand the use of placebo-focused trial designs (when feasible and ethically possible), such as the inclusion of no-treatment control groups (in addition to a placebo group), methods that ethically manipulate expectancy (eg, open-hidden paradigms), and optimisation of analytical methods that might correct for unequal distribution of placebo responses across trial groups; (3) formalise comprehensive critical appraisal of placebo groups in research training and guidelines; (4) require RCTs to report on blinding integrity; (5) develop neurophysiological biomarkers and objective measures from digital phenotyping technologies that might more objectively characterise and delineate placebo effects in

research settings; and (6) advocate reconceptualisation of inherent negative and stigmatised labelling of placebo effects that might stifle advancement and funding opportunities for these lines of research. The considerations presented in this review are broadly relevant across medical disciplines, and psychiatry should be a leader in developing new strategies for navigating and interpreting placebo effects in clinical research.

Contributors

All authors contributed to the conception, design, revision, and edit of this Review. MJB, JTT, BS, NDS, MA, TB, and CB drafted the manuscript. All authors had full access to the content of this manuscript and accept responsibility to submit for publication.

Declaration of interests

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